



DEFINITION

RISK

Uncertainty that matters



MATTERS?

- Health and safety
- Environmental
- Cost
- Schedule
- Stakeholder engagement

IMPACTS

ZERO HARM	PROJECT SUCCESS
• Eliminate any harm to people or the environment • Ethical responsibility	• Financial performance • Finishing on-time



PROJECT RISK MANAGEMENT

The processes of conducting risk management planning, identification, analysis, response planning and control on a project.



Increase the likelihood of positive events by exploiting opportunities while decreasing negative events by avoiding threats.

CATEGORIES OF RISKS

A background image of a construction site. In the foreground, two men wearing white hard hats and safety vests are standing on a concrete slab, looking at a tablet. The man on the left is wearing a grey jacket and a bright yellow safety vest. The man on the right is wearing a light blue shirt and dark pants. In the background, several other workers in safety gear are visible, along with a yellow crane and a building under construction. The scene is set on a high-rise construction site with many vertical rebar rods protruding from the concrete slabs.



Risk Categories - Categories of uncertainty that impact objectives (not objectives)

RISK CATEGORIES

Technical

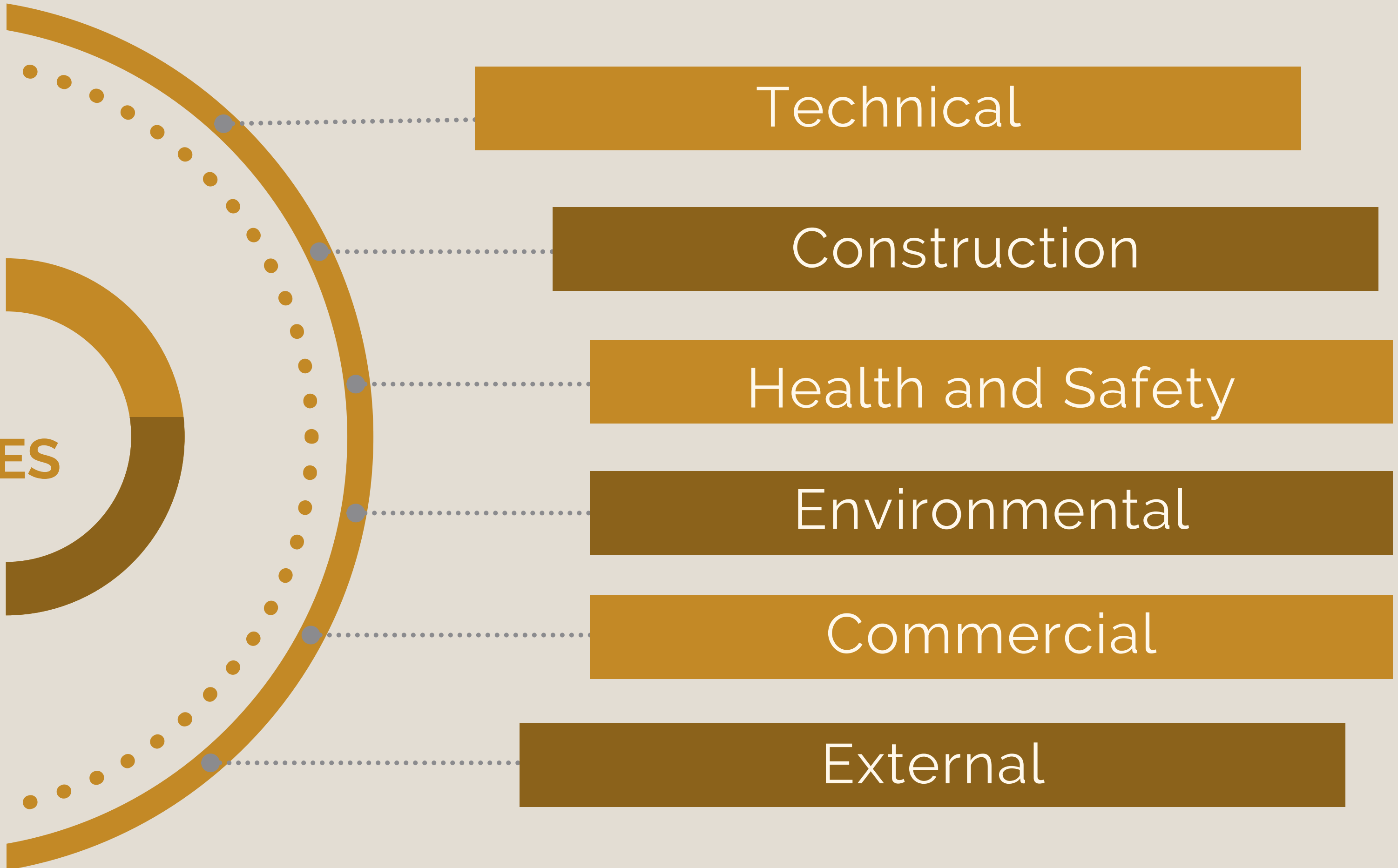
Construction

Health and Safety

Environmental

Commercial

External



TECHNICAL RISK

- Design risk
- Development of project technical solutions
- Ability to fulfill stakeholder requirements

I.e. Foundation strength meets earthquake ratings

HVAC equipment can control climate in building



CONSTRUCTION RISK

Expenditure of a significant amount of resources and dollars to deliver the design

I.e. Sub-contractor co-ordination

Procurement delays

Poor workmanship



HEALTH AND SAFETY

Risk of harm to workers, asset users,
and members of the public

I.e. Working at heights

Member of the public falls into worksite

Bridge collapses post-construction



ENVIRONMENTAL

Risk of harm to the environment

I.e. Oil spill

Polluted water runoff

Clearing native vegetation without approval



COMMERCIAL

Risks arising from commercial rights and obligations

I.e. Contract disputes

Late payment

Liquidated damages

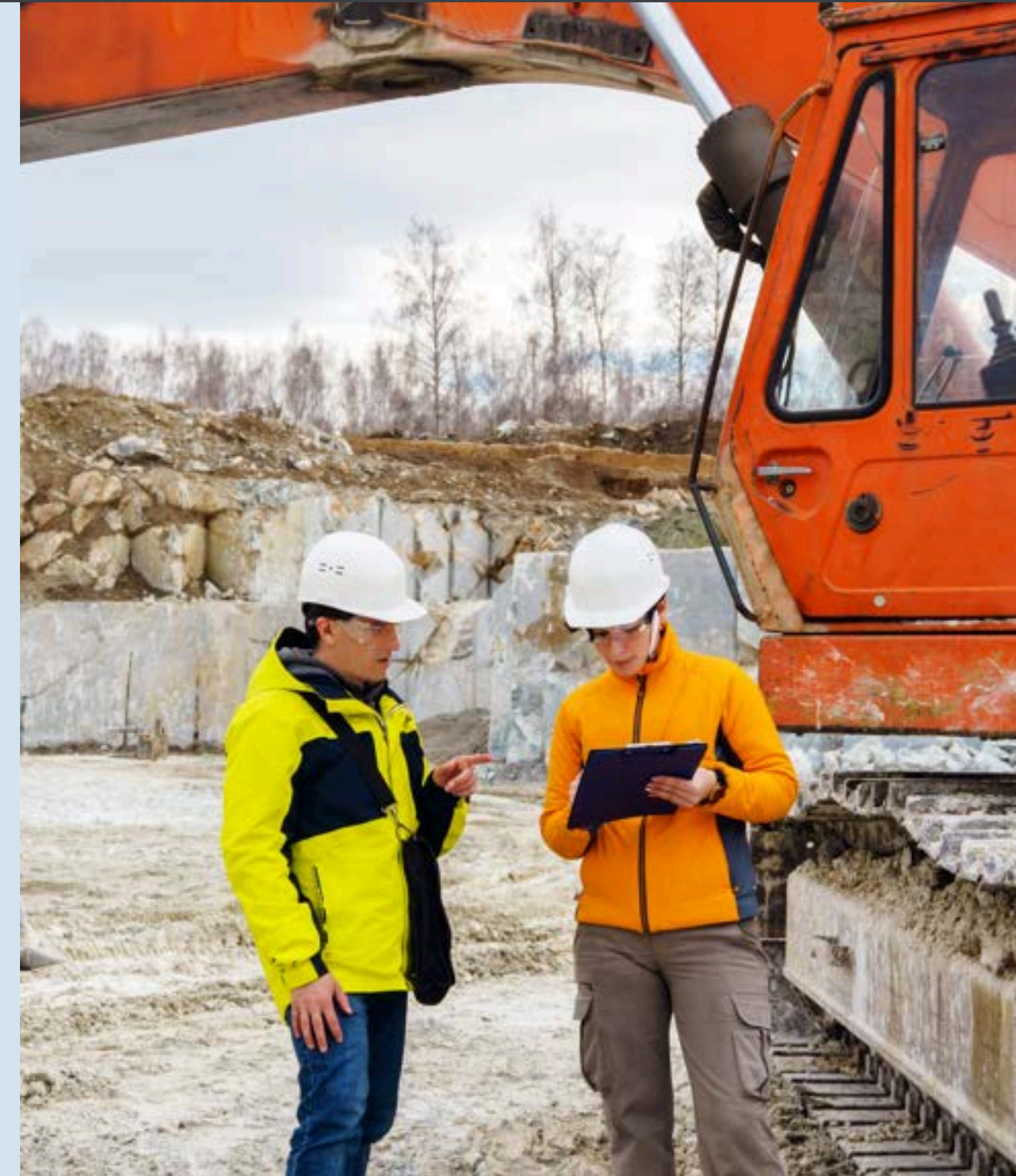


EXTERNAL

Risks arising from stakeholders
and impacted third parties

I.e. Unable to access worksite through
private property

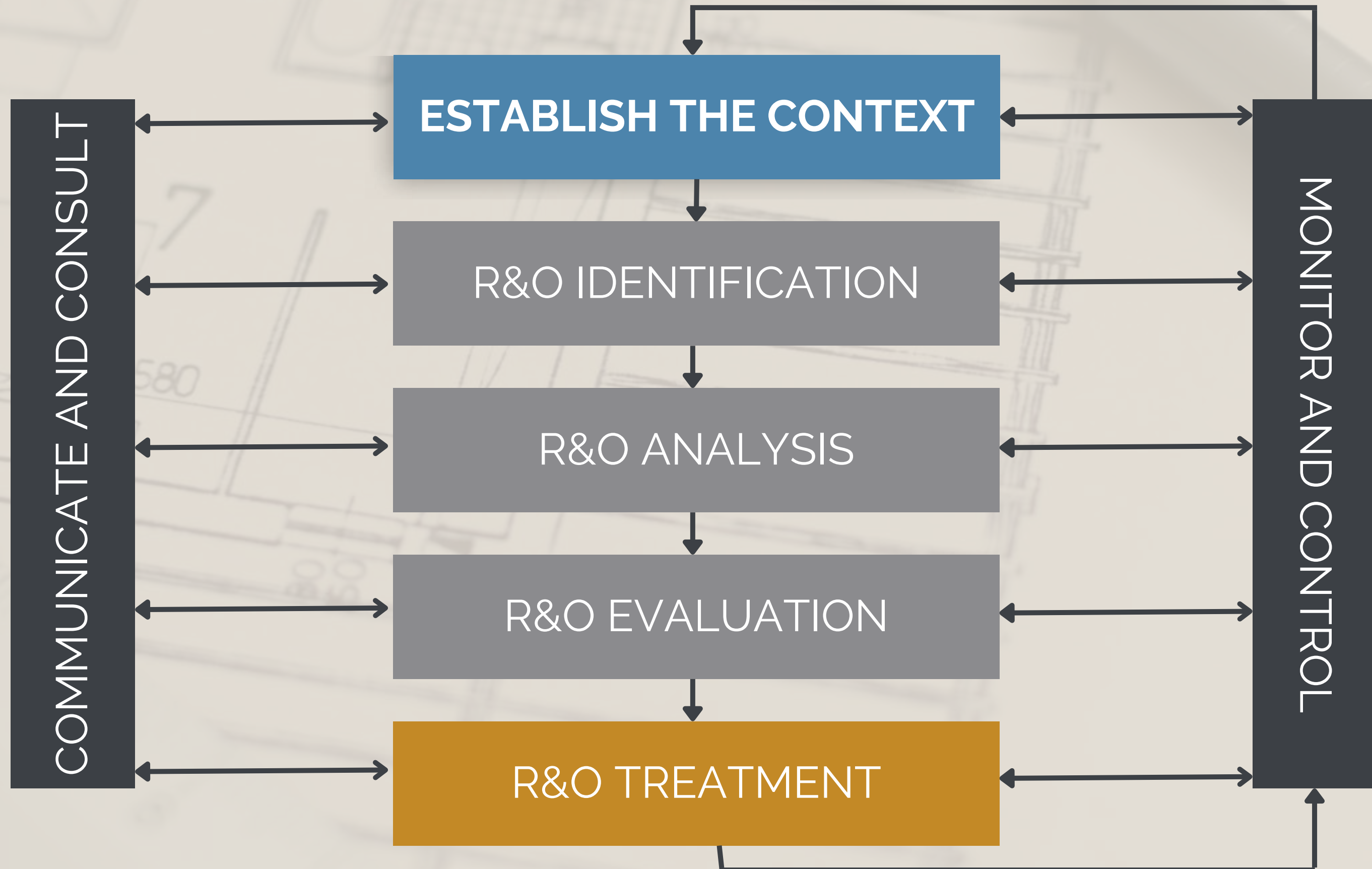
Utility not allowing works to proceed



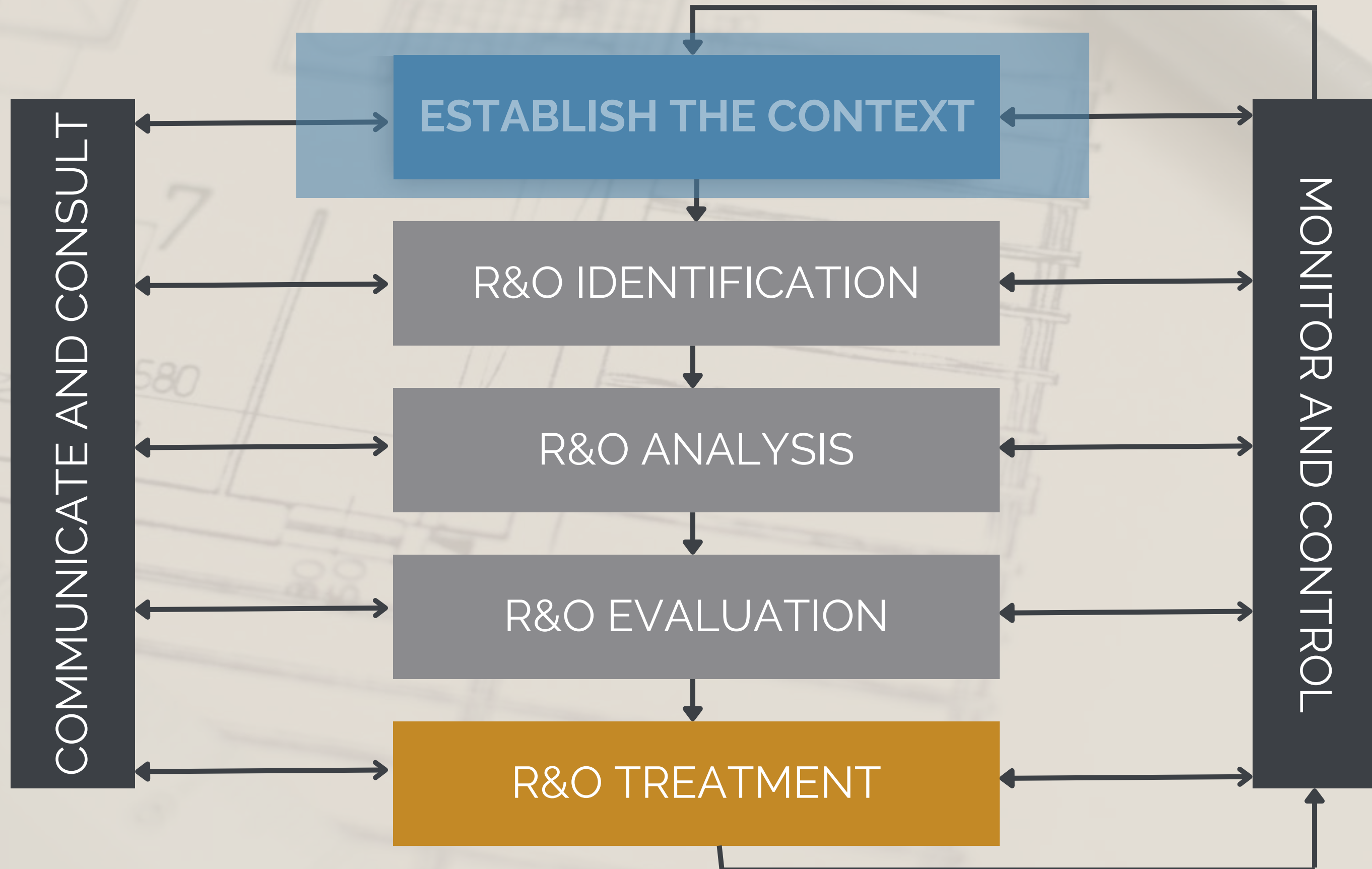
A background image showing three construction workers in safety gear (hard hats and high-visibility vests) standing on a construction site. One worker is pointing towards a large, brightly lit building under construction. The scene is set against a warm, orange-hued sky, suggesting sunset or sunrise. The workers are holding blueprints and a clipboard, engaged in a discussion.

RISK MANAGEMENT PROCESS

RISK MANAGEMENT PROCESS



RISK MANAGEMENT PROCESS



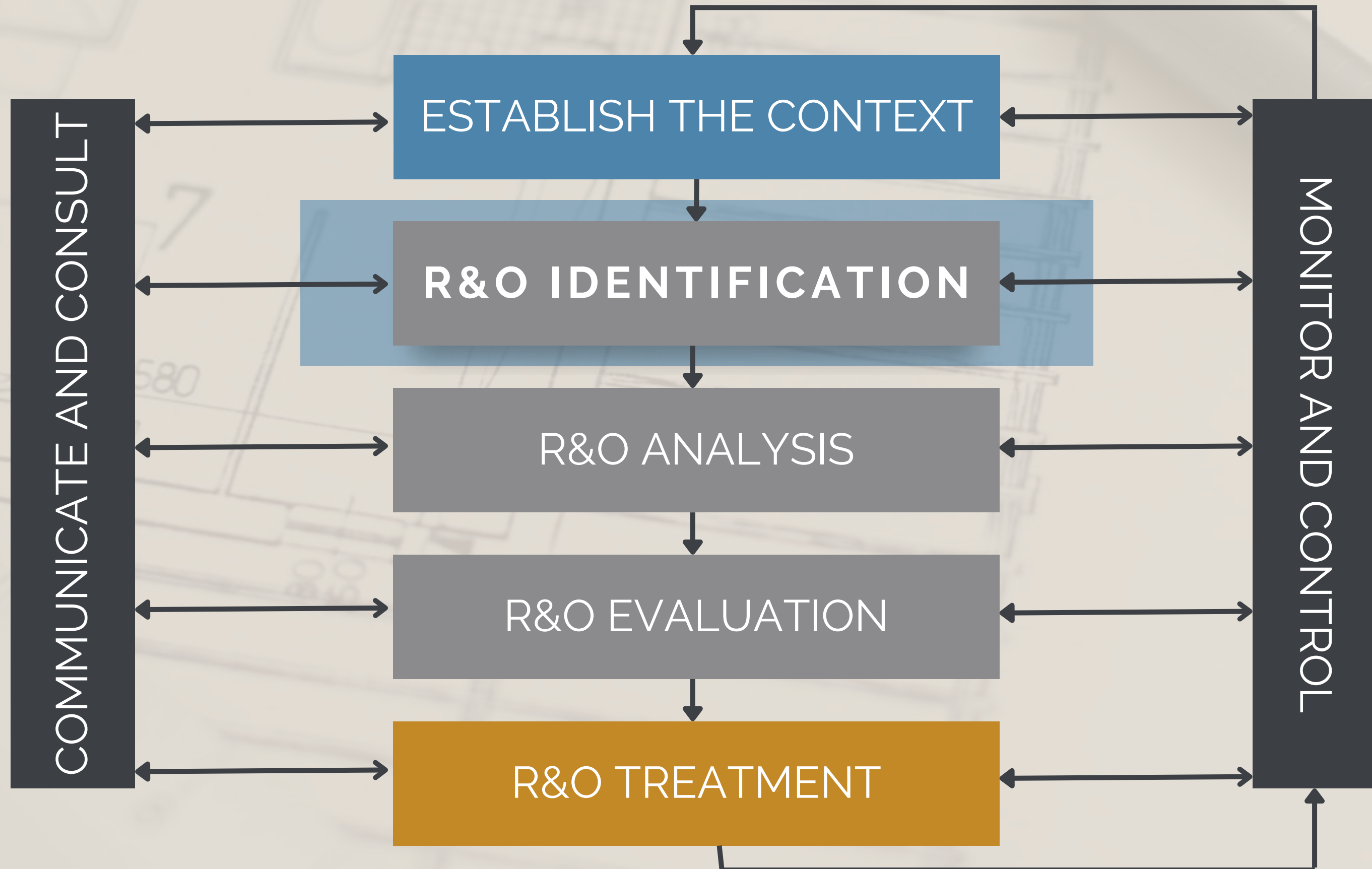
ESTABLISH THE CONTEXT

- Depth and complexity of risk responses
- Cost of risk management
 - Time
 - Resources
- Law of diminishing returns
- Tailor based on project specifics
- Risk thresholds

ESTABLISH THE CONTEXT - RISK THRESHOLDS

SCALE	SAFETY	SCHEDULE	BUDGET
LOW	Minor Injury	0-5 days	<\$5k
MEDIUM	Medium Injury	5-20 days	\$5k to \$30k
HIGH	Near Miss or Significant Injury	> 20 days	>\$30k

RISK MANAGEMENT PROCESS



RISK IDENTIFICATION

- Identify risks - not causes, effects, problems or issues

As a result of **cause**, **uncertain event** may occur, which would lead to **effect on objectives**

EXAMPLE RISK STATEMENTS

As a result of **latent site conditions**, **excess contaminated soil disposal** may occur which would lead to **increased disposal costs**.

01



RISK IDENTIFICATION TECHNIQUES

01 Past - what has happened on previous projects?

02 Present - what we are currently exposed to

03 Future- what could go wrong

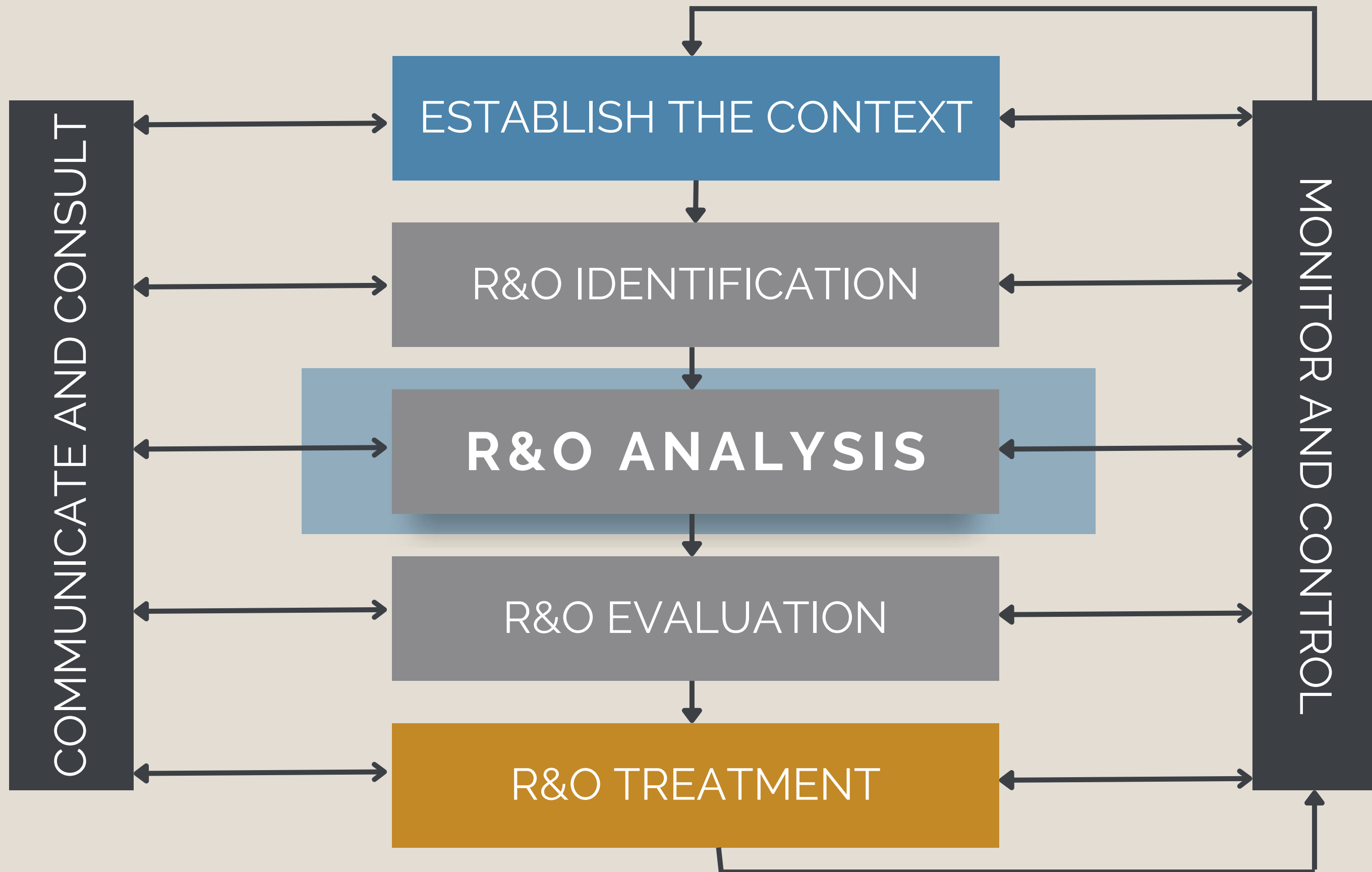
TECHNIQUES

- Lessons learned
- Subject matter experts
- Checklists
- Contract analysis
- Documentation analysis
- Brainstorming

RISK REGISTER

REF	NAME	CATEGORY	RISK DESCRIPTION	EFFECT
1	Excess contaminated Soil	Construction	Unkown quantities of contaminated soil on-site	Increased costs of disposal Delays to project schedule
2	Design Solution Failing to meet client requirements	Technical	Poorly understood client requirements may lead to design solution that client rejects	Delays to project schedule as design will not be approved
3	Scope growth	Construction	Client awarding us additional scope due to good project performance	Increased revenue and profit

RISK MANAGEMENT PROCESS



01. QUALITATIVE

Subjective risk assessment where likelihood and impact categorised into Low, Medium, High - used to rank risks



02. QUANTITATIVE

Uses verifiable data to assign a specific numerical data



QUALITATIVE RISK ANALYSIS

- Risk - Probability x Impact
- Probability - chance of occurring
- Impact - time, money, performance
- Two dimensions are assessed
 - Low, Medium, High
 - Different scales used

Risk Rating = Probability x Impact



RISK MATRIX

		Consequence				
		Negligible 1	Minor 2	Moderate 3	Major 4	Catastrophic 5
Likelihood	5 Almost certain	Moderate 5	High 10	Extreme 15	Extreme 20	Extreme 25
	4 Likely	Moderate 4	High 8	High 12	Extreme 16	Extreme 20
	3 Possible	Low 3	Moderate 6	High 9	High 12	Extreme 15
	2 Unlikely	Low 2	Moderate 4	Moderate 6	High 8	High 10
	1 Rare	Low 1	Low 2	Low 3	Moderate 4	Moderate 5

RISK REGISTER

Ref Number	Name	Impact Description	WBS Codes Impacted	Impact Level	Probability Level
1	Wet Weather	Excessive inclement weather delays construction works	All	3	3
2	Geotechnical	Excessive rock slows trenching productivities for installation of services	3.2.1	2	2
3	Design	Design growth post-tender resulting in increased construction costs	All	4	5
4	Scope Gaps	Scope gaps during procurement result in variations from sub-contractors	All (non self-perform)	4	2
5	Defects	Defects in self-perform works resulting in costly rectification works	3.2.1	3	3
6	Free-Issue Equipment Delays	Delays in free-issuing equipment to sub-contractors resulting in delay claims	3.2.3	1	1
7	Design	Value engineering opportunities that save construction costs	All	3	3
Total					

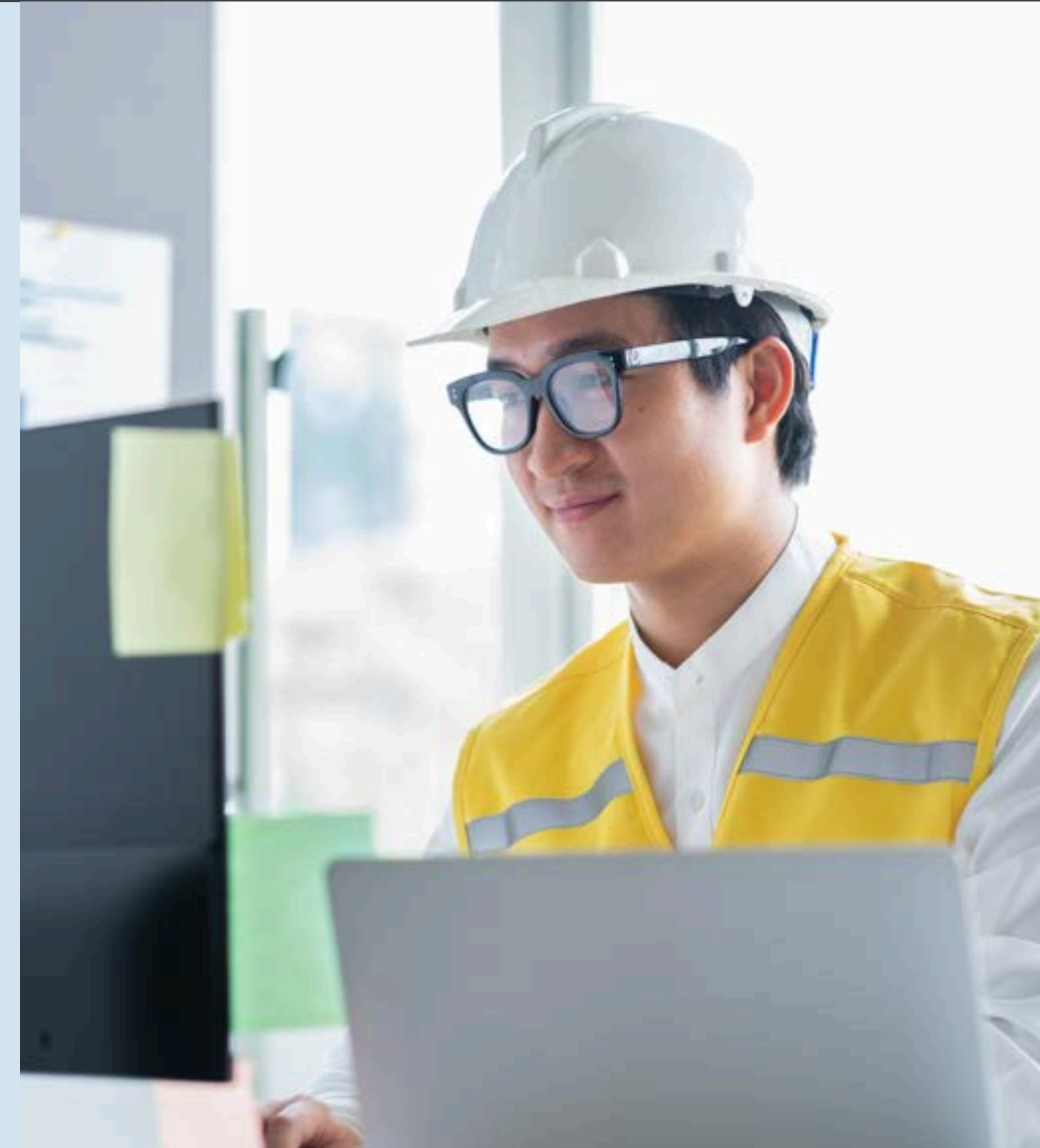
QUANTITATIVE RISK ANALYSIS

- Assigns a numerical value to risks
- Micro-level - looking at individual risks
- Macro-level - overall effect on project outcomes
 - Schedule
 - Budget

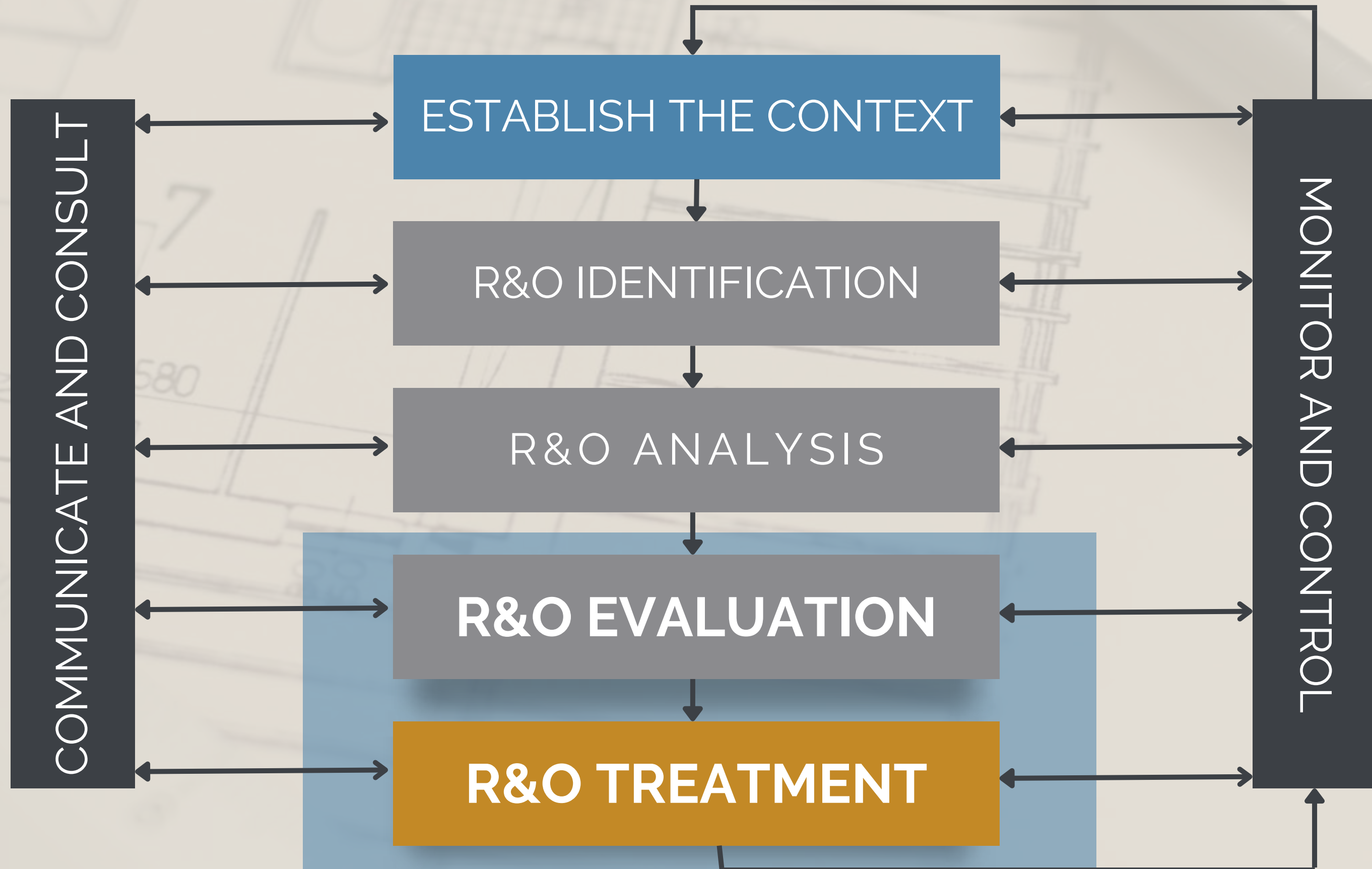


BENEFITS

- Determine the probability of achieving a specific project objective
 - I.e. P50 schedule
- Identify realistic project targets
- Exact values for risk events
- Analyse cost of risk responses



RISK MANAGEMENT PROCESS





WHAT CAN WE DO IN RESPONSE TO A RISK?

- Do nothing
- Treat the risk
- Undertake further analysis
- Maintain existing controls
- Reconsider objectives



Risk Treatment - taking some form of action in response to the risk to reduce the threat or maximize the opportunity.

AVOID/EXPLOIT

Eliminate the risk or make an opportunity definitely happen

TRANSFER/SHARE

Involve another person or party to share risk

TREATMENTS

REDUCE/ENHANCE

Reduction of the threat or enhance the opportunity

ACCEPT

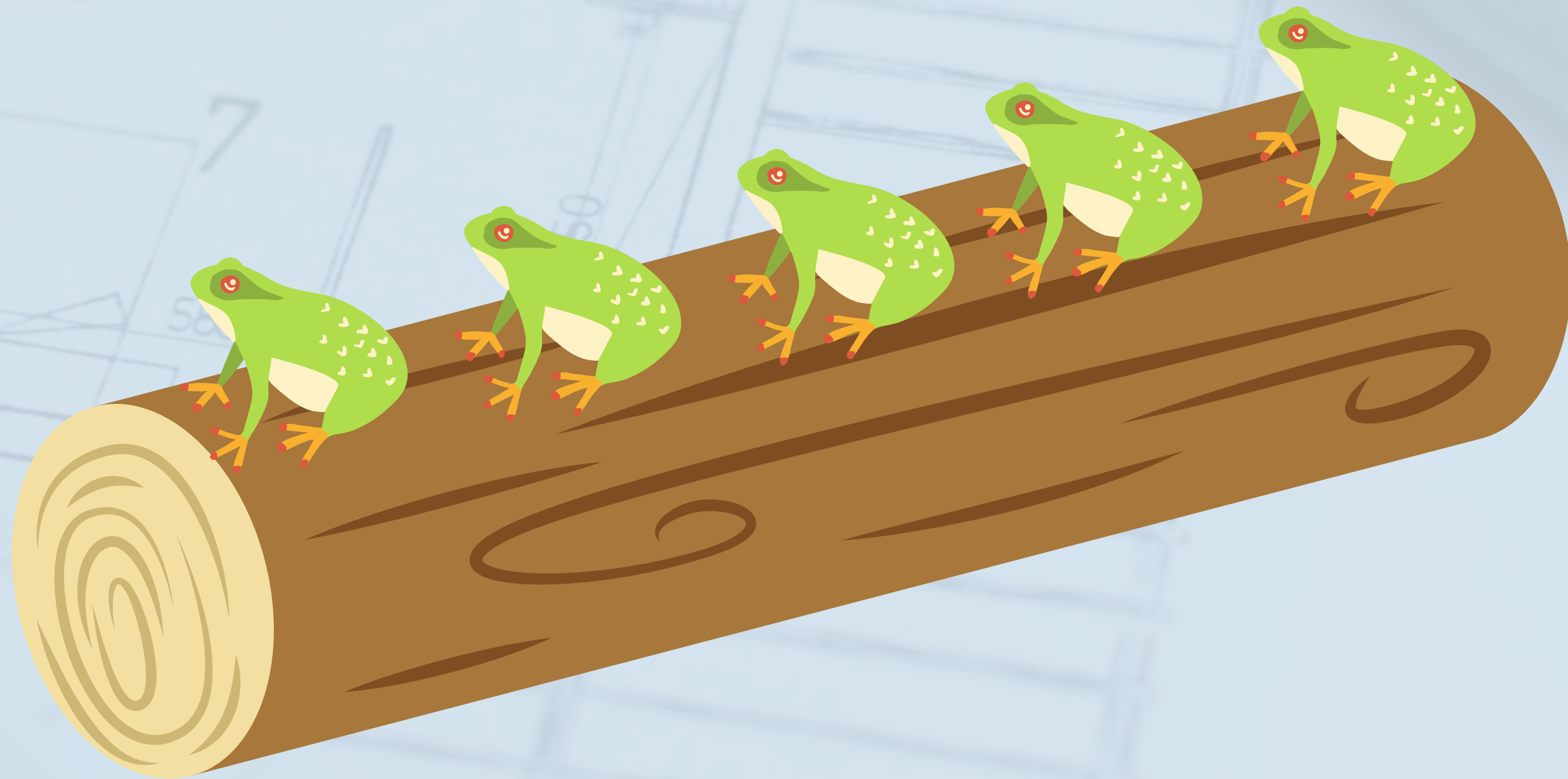
Take the risk without any special action



WHAT IS THE OPTIMAL STRATEGY?

- Specific risk
- Likelihood and impact
- Risk rating
- Urgency, manageability, propinquity, proximity
- Cost of risk response

HOW MANY FROGS ARE LEFT?





RISK RESPONSE IMPLEMENTATION

treat risk responses as project tasks - defined scope, duration, budget, completion criteria, and action owner



CONTAMINATED SOIL RISK

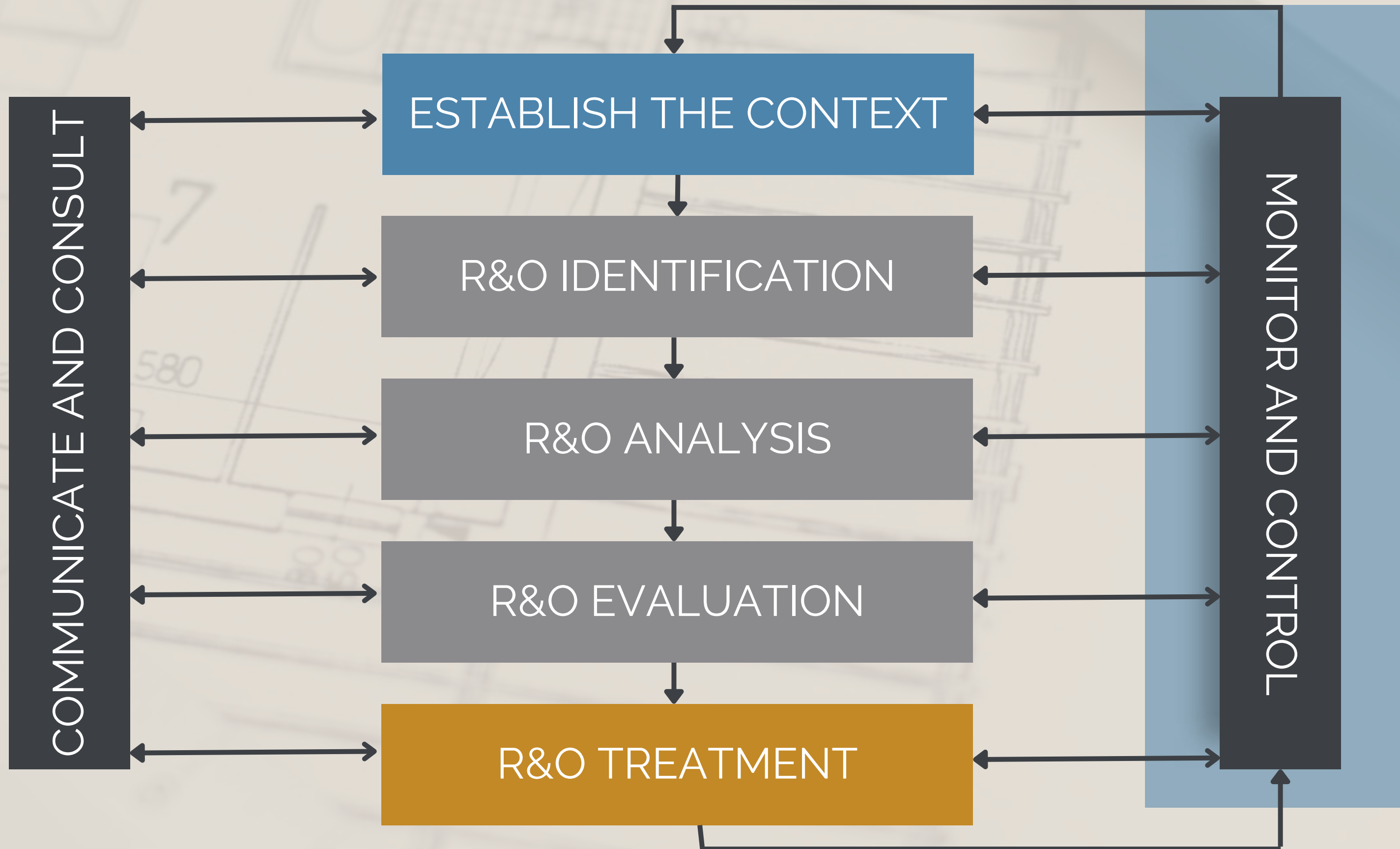
- Responsible for contaminated soil
- Risk response
 - Transfer to sub-contractor
 - **Timeline** - procurement phase
 - **Budget** - \$30k
 - **Responsible** - earthworks package lead

CONTROLLED RESPONSE

- Clear how plan will be implemented
- Known deadline for implementation
- Re-assess if plan not followed
- Known budget
- Responsible person
- Change management



RISK MANAGEMENT PROCESS

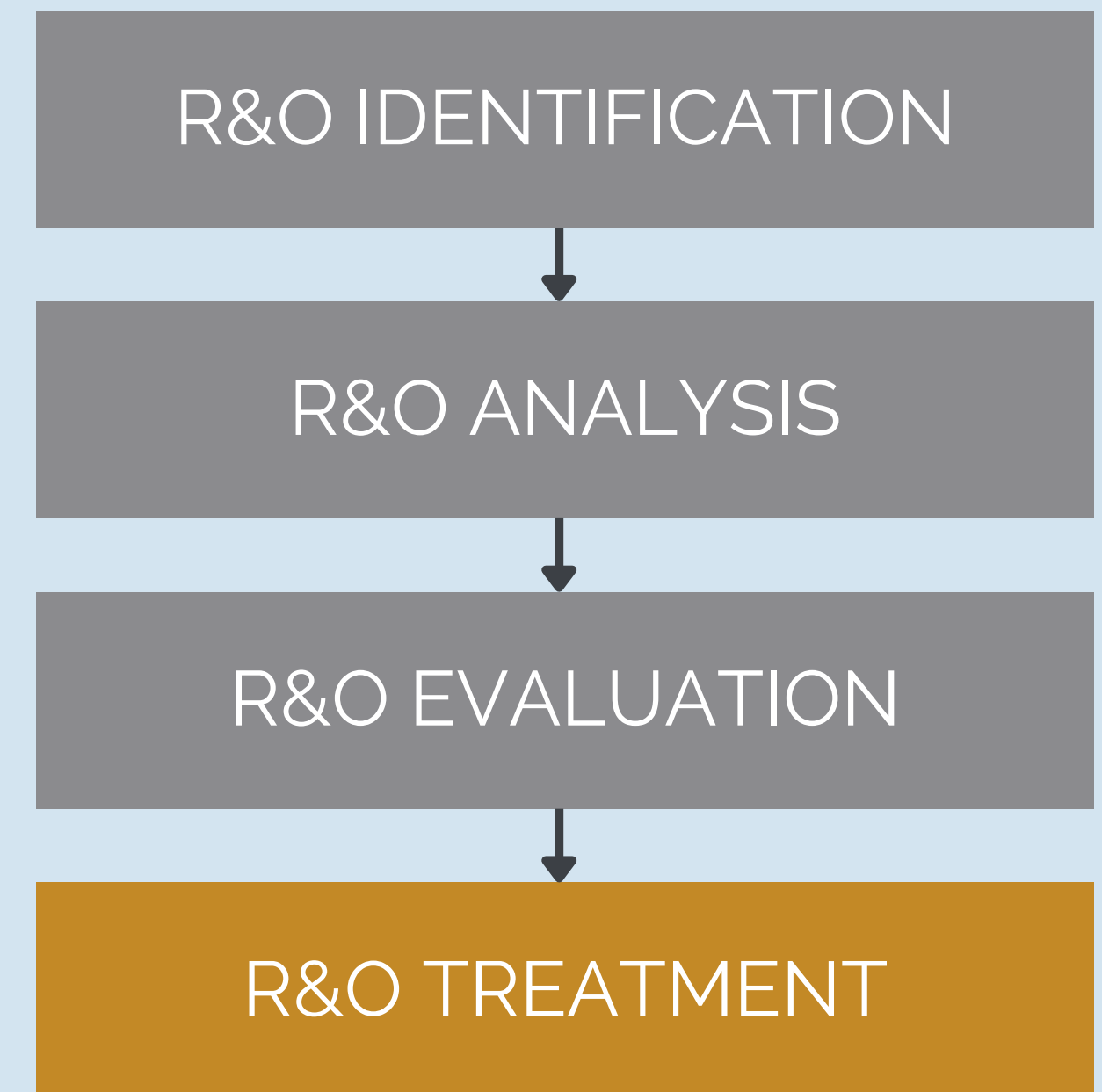


**MONITORING AND
CONTROLLING RISK**



RISK PROCESS ITERATIONS

- Regularly re-doing the risk management process
 - I.e. Monthly
- Identify and treat emergent risks
- Current status of project risk level, contingency, etc.



SUMMARY

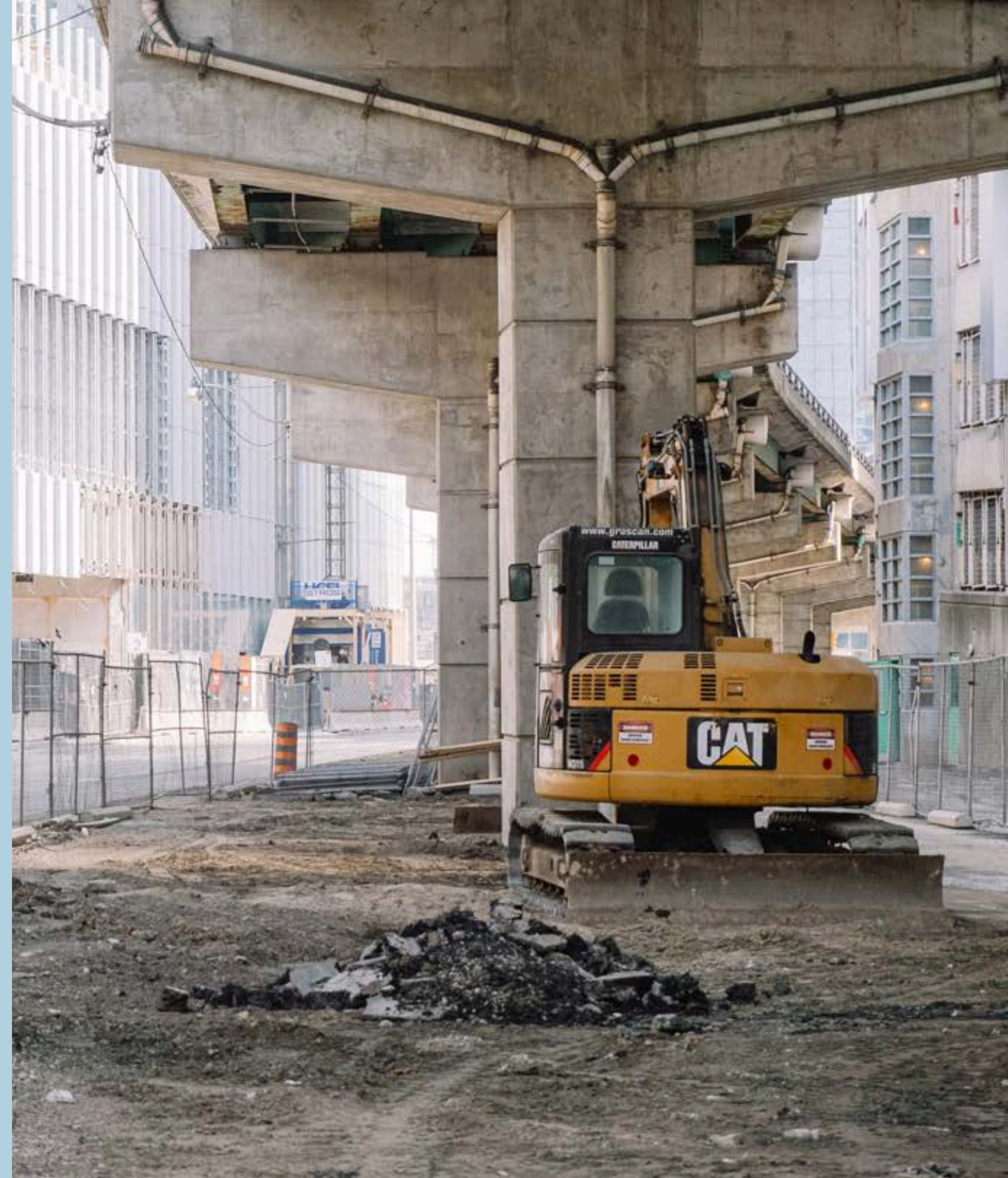
- Risk is the uncertainty that matters to the project
- Risk management - eliminating threats and exploiting opportunities
- Series of steps we follow
 - At the beginning of the project
 - Iteratively
- Risk is a reality of projects

WHAT HAZARDS CAN YOU SEE?



HAZARDS

- Traffic
- Underground services
- People and plant
- Dropped objects
- Structure collapse
- Machinery failure
- Trips
- Oil spill



HSE RISK MANAGEMENT - ENSURING ITS SAFE





HEALTH AND SAFETY RISK MANAGEMENT

The process of managing construction activities to eliminate or minimize as far as reasonably possible the risk of injuries or fatalities.

EXAMPLE: WORKING AT HEIGHTS

Use of scaffolding and
harnesses to safely
work at heights



EXAMPLE - ELECTRICAL SAFETY

Routine inspections on temporary electrical supplies to ensure they are in safe and good working order



EXAMPLE - TOOLBOX TALKS

Regular toolbox talks with the workforce to ensure focus on safety and critical risks maintained



The fundamental ethical responsibility of a project manager





**If the job can't be done
safely, then it can't be done.**



**How do we know if a job can
be done safely?**

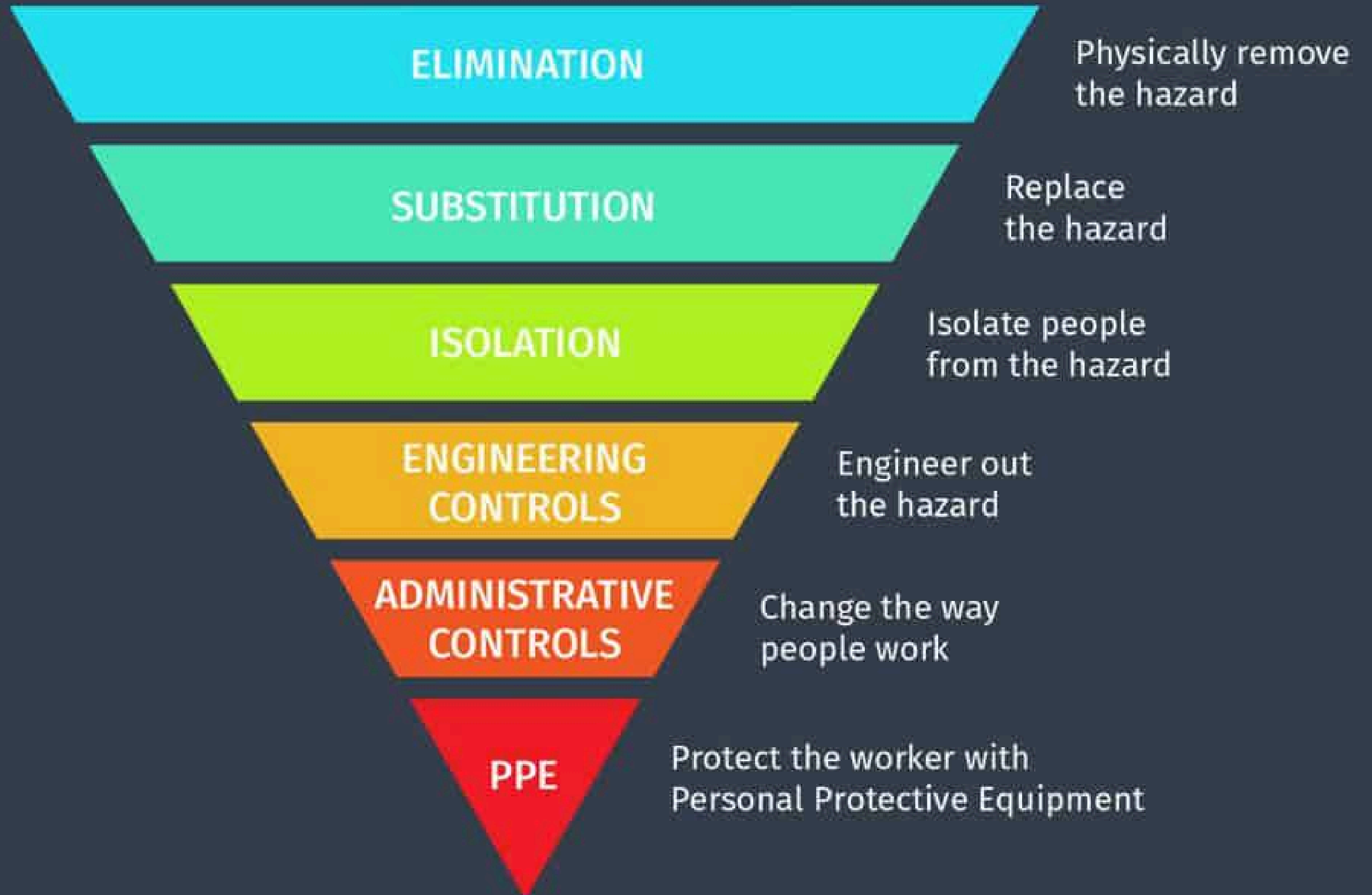
A photograph of three construction workers on a site at sunset. The workers are silhouetted against a bright orange and yellow sky. They are wearing hard hats and high-visibility vests. The worker on the left is gesturing with his right hand while holding a rolled-up blueprint. The worker in the middle is holding a large sheet of paper. The worker on the right is holding a clipboard. In the background, there are blurred lights from a building and some construction equipment.

**Tha hazards have
appropriate controls in place**

MOST EFFECTIVE



LEAST EFFECTIVE





ELIMINATION

- Physical removal of the hazard
- Effective during design/planning
- E.g. complete activity at ground level rather than at heights



SUBSTITUTION

- Substitute hazards with a safer option
- E.g. excavation with vacuum excavation rather than an excavator



ISOLATION

- Physically isolate people from the hazard
- E.g. concrete barriers to protect workers from road traffic



ENGINEERING CONTROL

- Put in place a system to engineer out the risk

I.e. Edge protection



ADMINISTRATIVE

- Put in place a system to engineer out the risk
- E.g. using traffic signs or cones to warn people



PPE

- Personal Protective Equipment (Helmets, gloves, Hi-Vis, glasses)
- Least effective form

MOST EFFECTIVE

ELIMINATION

Physically remove
the hazard

SUBSTITUTION

Replace
the hazard

ISOLATION

Isolate people
from the hazard

ENGINEERING
CONTROLS

Engineer out
the hazard

ADMINISTRATIVE
CONTROLS

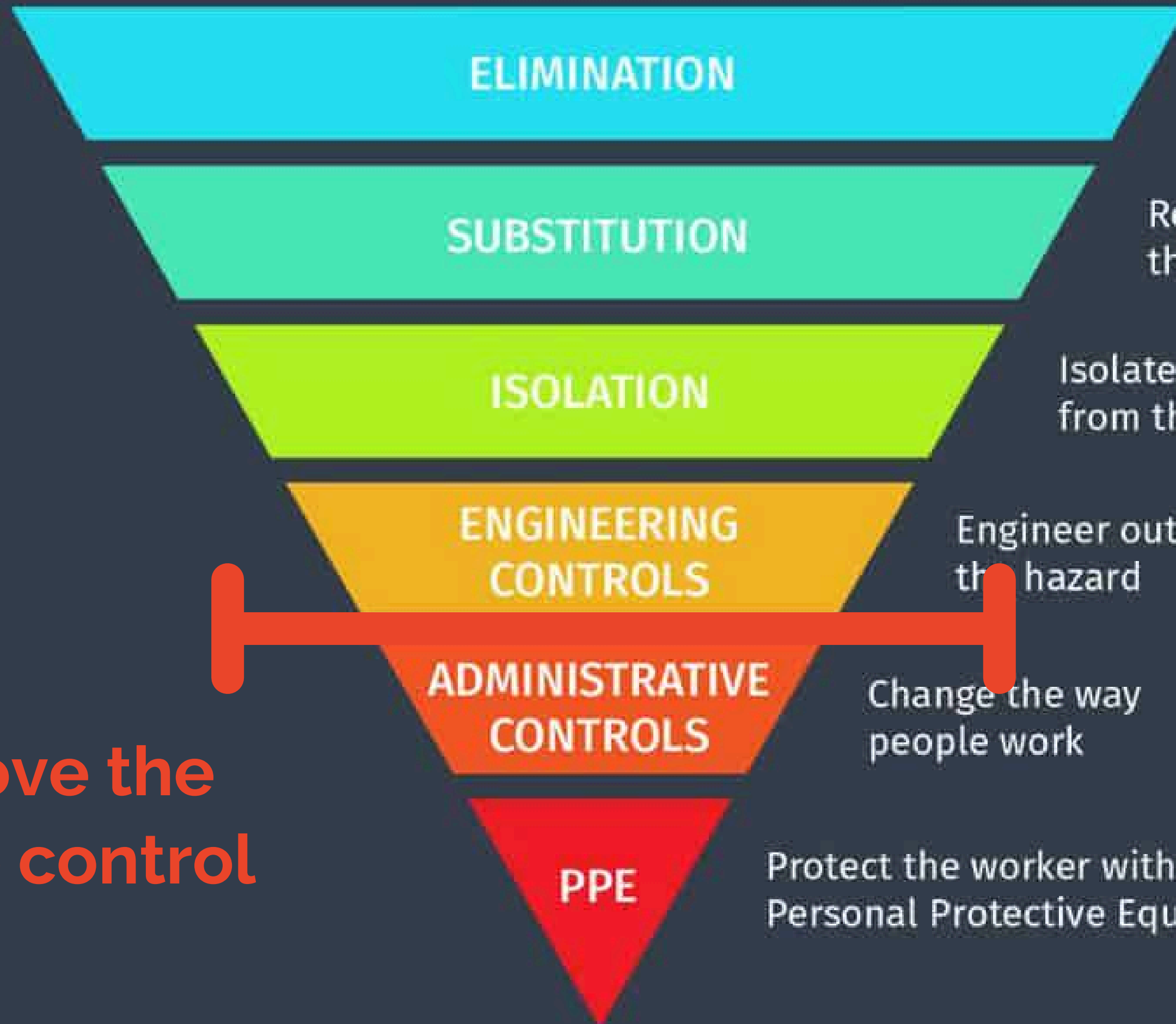
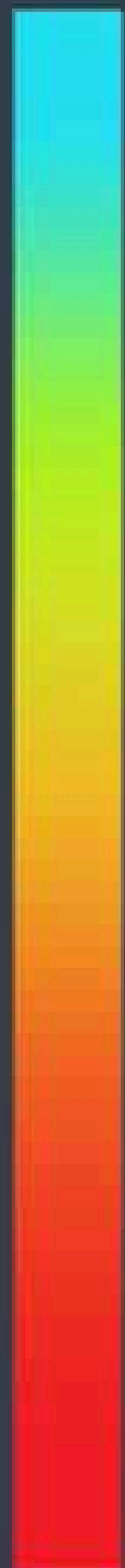
Change the way
people work

PPE

Protect the worker with
Personal Protective Equipment

Above the
line control

LEAST EFFECTIVE



WHAT IS SAFE ENOUGH?

- Bare minimum acceptable are engineering controls
- Engineer out the hazard
 - Guards on machines
 - Ventilation in confined spaces
 - Edge protection when working at heights
- PPE/Administrative controls are not acceptable
 - “You can’t go work next to that exposed edge”
 - “Wearing a helmet will protect you if you fall”

A construction worker is shown from the side, wearing a white hard hat, a black long-sleeved shirt, and an orange safety harness. The worker is pulling a rope upwards with their right hand. The background is a construction site at sunset, with several cranes and building structures visible against a warm, orange and yellow sky. The sun is low on the horizon, creating a strong backlight effect.

What tools do we have?

SAFE WORK METHOD STATEMENTS (SWMS)

- Document used to identify and control all hazards
- Minimum controls in place
- Needed for all high-risk construction activities



SAFE WORK METHOD STATEMENT

What are the tasks involved?	What are the hazards and risks?	What are the risk control measures?		
List the HRCW work tasks	List the hazards and risks of the task	List the risk control measures	List how the control measures will be implemented	List who is responsible for the control measure being implemented
EXAMPLE: • Roof tiling	EXAMPLE: • Slipping or falling from the roof	EXAMPLE: • Scaffold with catch platform and or guard rail system • Fall restraint system such as harness and appropriate anchor point	EXAMPLE: • Scaffold or guard rail supplied and erected by supplier or competent person • Fall restraint system installed and used by appropriately trained persons	EXAMPLE: • Principal contractor/builder • Roofer/roofing supervisor

Safety in Design (SiD)

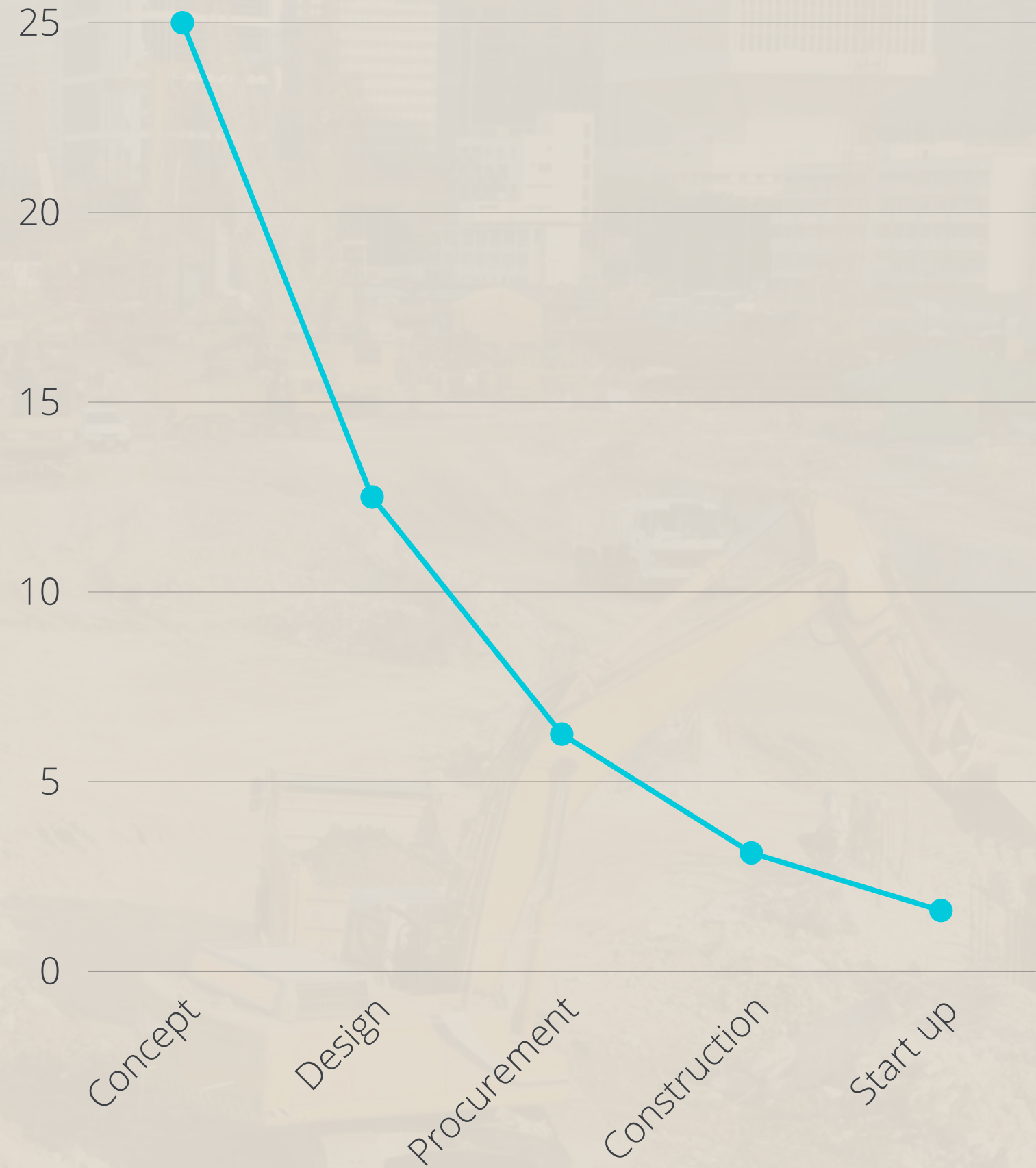
- Series of workshops during the design phase with representatives from all phases
- Comments raised and recorded in SiD register
- Comments closed out during design process
- Above-the-line controls
- Legislative requirement



INFLUENCE ON SAFETY OVER TIME

As the project progresses from planning into delivery, our ability to influence safety decreases

ABILITY TO INFLUENCE CHANGE



SAFETY IN DESIGN EXAMPLE

[illegible]

Competency/Training

- Verify workers have the skills and expertise to complete tasks
 - Electricians performing electrical works
 - Excavator operator is competent
- Project specific
 - Site inductions
 - Industry standards (Rail)
 - Job/task specific



Legislation

- Failure to provide a safe workplace is criminal offense
 - Different from contract law
 - Jail time/large fines as punishment
- OH&S mandated by legislation
 - SWMS
 - Safety in Design
 - Specific risks (trenches deeper than 1.5m)



Awareness

- Ensuring safety is at the forefront of everybody's minds
- Workplace culture with a focus on safety
- Examples
 - Pre-starts
 - Toolbox talks
 - Company branding



Reporting

- Reporting on key health and safety metrics
- Accountability and responsibility
- Lead indicators and trend analysis
- Examples
 - LTIs
 - TRIFR



KEY TAKEAWAYS

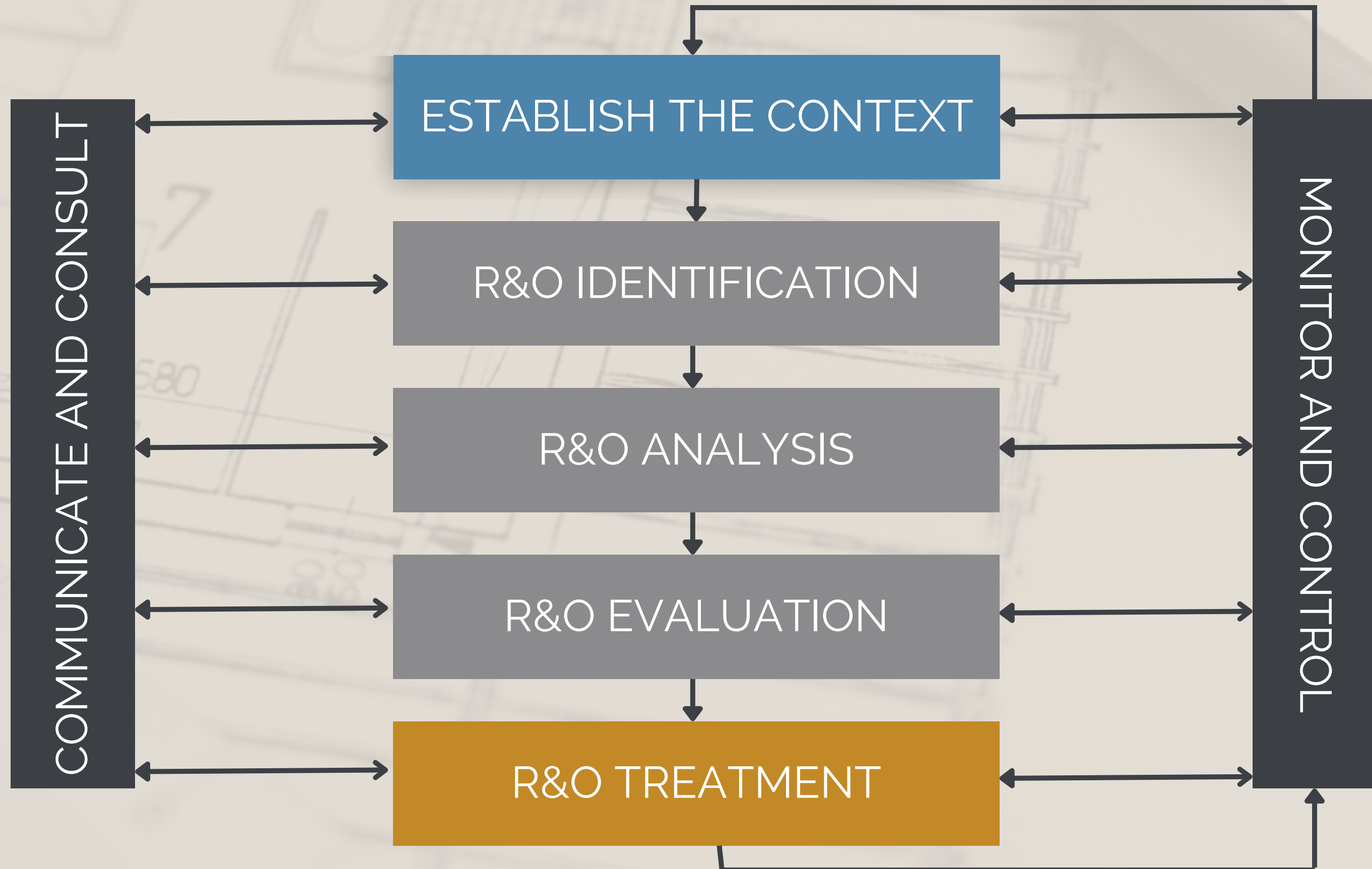
- Safety is an ethical responsibility, not a buerocratic activity
- If we can't do something safely, we don't do it
- What is "safe" enough?
 - Minimum of an engineering control in place
- Variety of tools available at our disposal to control hazards

PROJECT

- Construction of a new wind farm
- Location - Coastal area with high wind potential
- Objective - generate electricity, connect to local power grid



RISK MANAGEMENT PROCESS



STEP 1 - ESTABLISH THE CONTEXT

- **Project Overview:** Construction of a wind farm in a remote coastal area to generate renewable energy and connect to the local power grid.
- **Project Goals:** Timely installation and activation of turbines to meet energy production targets and capitalize on favourable weather conditions.
- **Stakeholders:** Project owners, contractors, logistics providers, local authorities, and environmental agencies.
- **Constraints:** Challenges include complex logistics, coastal weather variability, and environmental compliance requirements.

STEP 1 - ESTABLISH THE CONTEXT

- Risk thresholds
- Impact
 - Low - 1%
 - Medium 3-5%
 - High >5%
- Likelihood
 - Low - 10%
 - Medium - 10% to 50%
 - High >50%

Risk Matrix		Severity				
		Insignificant	Minor	Moderate	Major	Severe
Likelihood	Almost Certain	Medium	High	Very High	Very High	Very High
	Likely	Medium	High	High	Very High	Very High
	Possible	Low	Medium	High	High	Very High
	Unlikely	Low	Low	Medium	Medium	High
	Rare	Low	Low	Low	Low	Medium

STEP 2 - IDENTIFICATION

- **Stakeholder Engagement:** Conducted interviews with key stakeholders (logistics providers, regulatory bodies, and contractors)
- **Risk Workshop:** workshop with the project team to brainstorm risks and opportunities
- **Documentation review**
 - EPC contract
 - Project specifications
 - Grid connection requirements
 - Environmental approvals

STEP 2 - IDENTIFICATION

- **Weather Interruptions:** High winds or storms potentially halting construction.
- **Supply Chain Delays:** Delays in receiving turbine components (e.g., blades, nacelles).
- **Transportation Challenges:** Difficulties in moving oversized components to the remote coastal site.
- **Technical Failures:** Risk of malfunction or underperformance of turbines.
- **Environmental Compliance:** Ensuring adherence to wildlife and environmental regulations.

STEP 3 - R&O ANALYSIS

- **Focused Risk:** Transportation Challenges – Delays or issues in transporting large turbine components to the site.
- **Likelihood:** High (due to remote location and route complexity).
- **Impact:** High (could significantly disrupt critical path tasks).
- **Risk Rating:** Critical – requires a detailed risk management plan.

○



STEP 4 - R&O EVALUATION

- Assess each risk based on impact on schedule, budget and stakeholder expectation
- Risk prioritization
 - Transportation Challenges: High priority due to its direct impact on the timeline and dependencies.
 - Weather Interruptions: Moderate priority; requires continuous monitoring but minimal control.
 - Supply Chain Delays: Moderate priority with potential mitigation options.

STEP 5 - R&O TREATMENT

- Transportation challenges
- **Route Planning and Permits:** Work with local authorities to secure necessary permits and plan the best routes for oversized loads.
- **Engage Specialized Transport Providers:** Partner with logistics experts experienced in moving large wind components.
- **Staging Area:** Set up a local staging area near the site to store components, ensuring they're ready for installation when needed.
- **Weather Considerations:** Plan transportation during low-risk weather periods to reduce delays.

○

STEP 6 - R&O MONITOR

- **Real-Time Tracking:** Use GPS and regular check-ins to track component deliveries and monitor transport schedules.
- **Risk Review Meetings:** Weekly risk reviews to assess current risks and adjust treatment plans if new issues arise.
- **Contingency Adjustments:** Adjust transport schedules and use the staging area if transportation delays or adverse weather conditions arise.



SUMMARY

- Proactive planning is a form of risk management
- Risk management process comes down to effort and action
- Focus on what you can control and take action